Name:_____

Exam Question Booklets
OCR Physics A

Booklet 4
13 Questions
106 marks

Things to remember:

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1 An apple falls onto the head of a philosopher sitting beneath an apple tree.

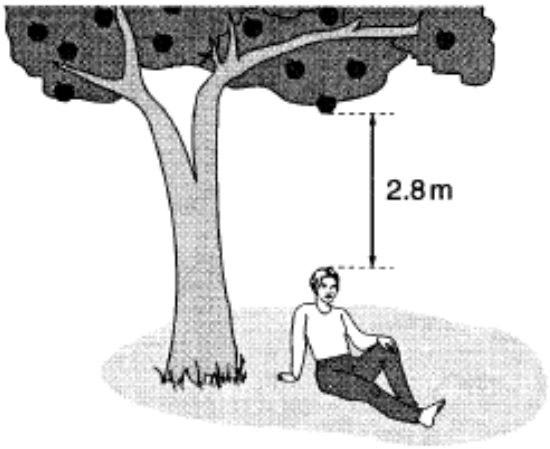

Fig. 2.1

The apple of mass 0.15 kg falls from rest through a vertical height of 2.8 m before striking the philosopher.

(a) Describe the energy change taking place as the apple falls.

[1]

(b) **(i)** Calculate the speed at which the apple is travelling when it strikes the philosopher.

$g = 9.8\ \text{N kg}^{-1}$

speed = ... m s^{-1} [2]

(ii) State an assumption you have made.

[1]

Fig. 2.1 shows apparatus used to determine the acceleration due to gravity.

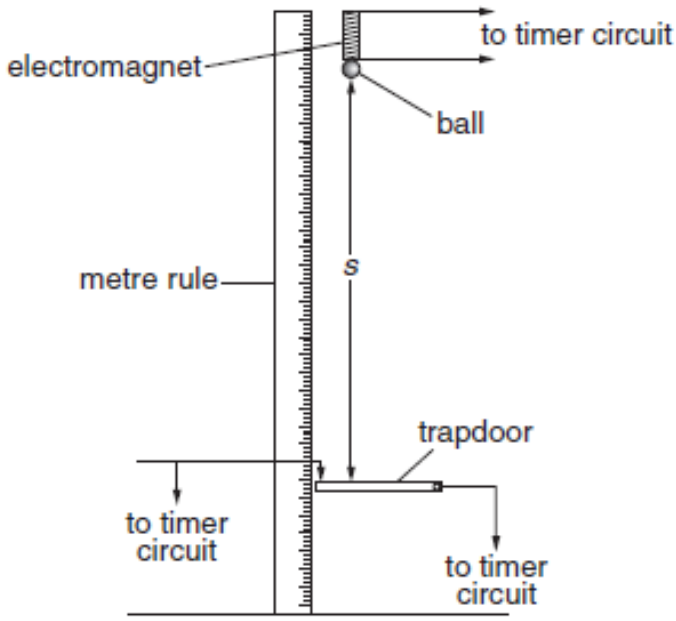

Fig. 2.1

A clock starts when a ball is released from an electromagnet and stops when the ball opens a trapdoor. The clock then displays the time t for the ball to fall a distance s. The trapdoor is moved to change the value of s. Fig. 2.2 shows the variation of t^2 against distance s.

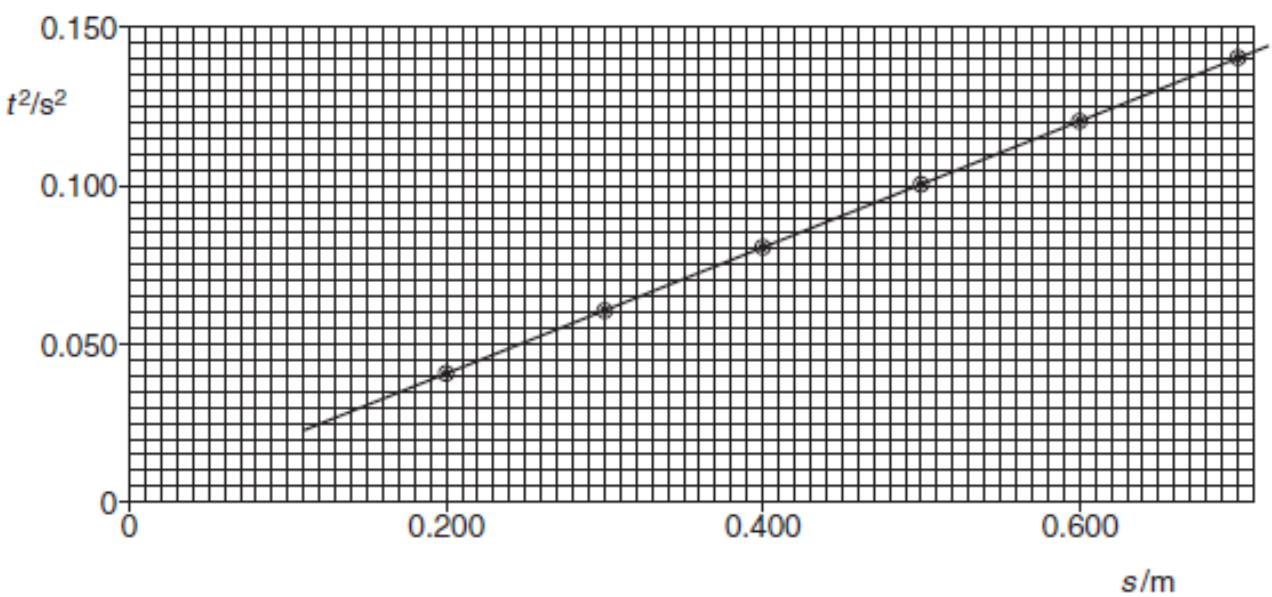

Fig. 2.2

(a) Use an equation of motion to explain why the graph in Fig. 2.2 is a straight line.

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.....

..... **[2]**

(b) Use Fig. 2.2 to calculate

(i) the gradient of the graph

gradient = ... **[2]**

(ii) the acceleration of the ball.

acceleration = ... $m\,s^{-2}$ **[2]**

(c) When a student performs this experiment he obtains a straight line graph which does not pass through the origin. Suggest **two** reasons why.

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.....

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..... **[2]**

3 Fig. 2.1 shows a trolley of mass 0.80 kg, on a bench surface, connected to a mass M by a string. The mass M is released and the trolley moves along the surface. Fig. 2.2 shows the variation of the velocity v of the trolley with time t for the motion from A to B.

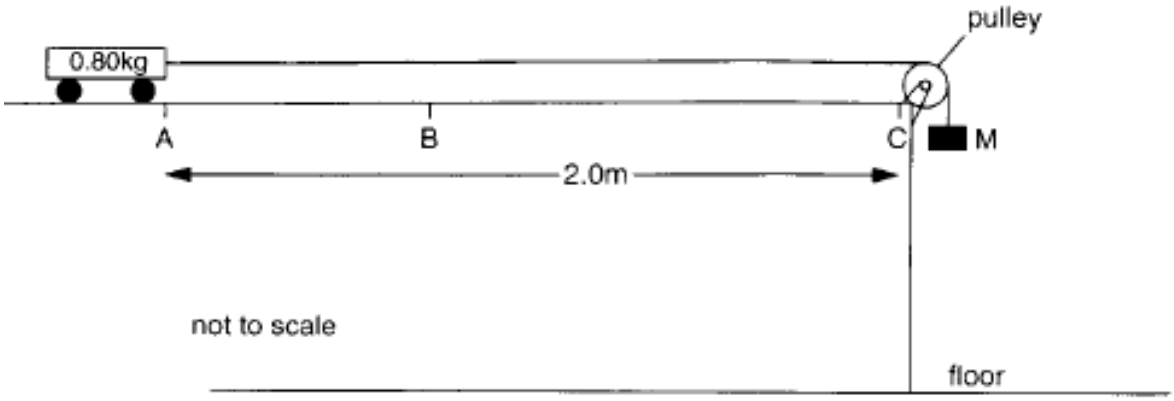

Fig. 2.1

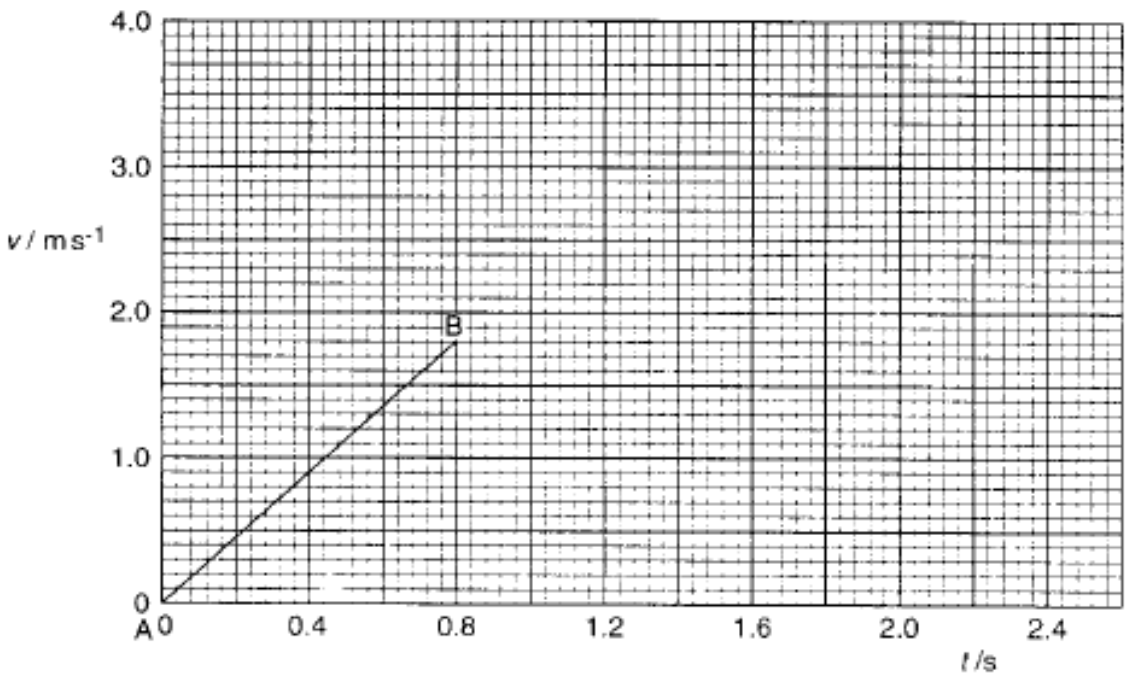

Fig. 2.2

(a) **(i)** Calculate the acceleration of the trolley between A and B.

acceleration = ... $m\,s^{-2}$ [2]

(ii) Calculate the resultant force acting on the trolley between A and B.

force = ... N [2]

(iii) Show that the distance from A to B is 0.72 m.

[2]

(b) When the trolley reaches B the mass M has just reached the floor.

(i) Ignoring any resistive forces, calculate the time it takes the trolley to travel from B to C.

time for B to C = ... s [3]

(ii) On Fig. 2.2, complete the graph for the trolley moving from B and coming to rest at the pulley at C. [3]

4 A skydiver of mass 75 kg jumps from an aircraft that is in level flight at a great height above the earth's surface.

(a) Calculate the gravitational pull of the earth on the skydiver as he falls from the aircraft.

Assume that $g = 9.8\,\text{Nkg}^{-1}$ at this height.

gravitational pull = ... N

[2]

(b) **(i)** Initially, the skydiver falls faster and faster, accelerating downwards.

Explain, in terms of the **forces** acting, why the downward acceleration of the skydiver *decreases* as his vertical speed *increases*.

[3]

(ii) Only a few seconds after leaving the aircraft the skydiver is falling towards the earth at a steady maximum speed of about $50\,\text{m}\,\text{s}^{-1}$ (over 100 miles an hour).

Explain why he reaches a *steady* speed.

[1]

(c) To land safely the skydiver opens his parachute. When he does this, his vertical speed decreases *rapidly* from $50\,m\,s^{-1}$ to about $5\,m\,s^{-1}$. He then falls to the earth at a steady speed of about $5\,m\,s^{-1}$.

Explain, in terms of the forces acting, why he

(i) decelerates rapidly when the parachute opens

[2]

(ii) falls with a smaller steady speed than before the parachute was opened.

[2]

Fig. 1.1 shows the path of a ball thrown from **A** and passing through positions **B**, **C** and **D**.

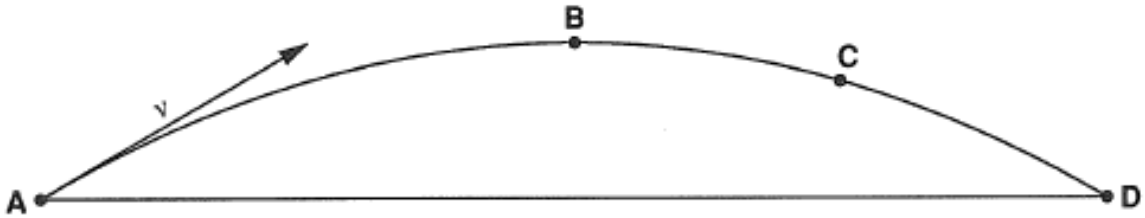

Fig. 1.1

The ball is thrown from **A** with a velocity v. A vector arrow on Fig. 1.1 represents the magnitude and direction of the velocity of the ball at **A**.

(a) On Fig. 1.1 draw arrows to represent the horizontal and vertical components of the velocity of the ball at **A**. [1]

(b) State how the components of the velocity of the ball at **B**, **C** and **D** compare with the components at **A**. Assume air resistance is negligible.

(i) The vertical component at **B** .....

The horizontal component at **B** .....[1]

(ii) The vertical component at **C** .....

The horizontal component at **C** .....[1]

(iii) The vertical component at **D** .....

The horizontal component at **D** .....[1]

(c) Explain the answers you have given for the components of the velocity of the ball at positions **B**, **C** and **D**.

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.....[4]

(a) A stone is projected horizontally from a cliff. Fig. 3.1 shows the stone at a position **A** on its path.

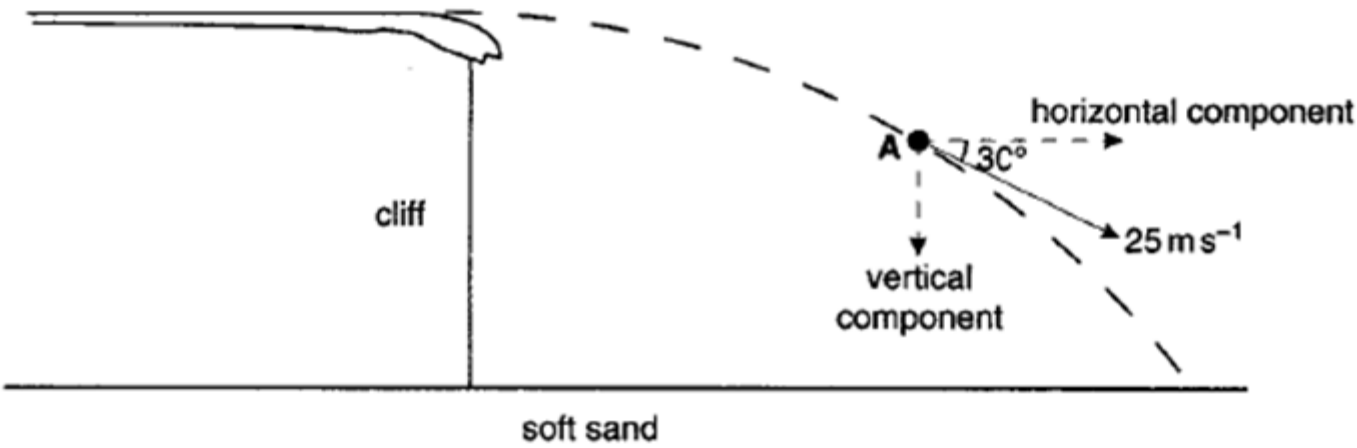


Fig. 3.1

The velocity of the stone at position **A** is 25 m s^{-1} at 30° to the horizontal.

(i) Show that the horizontal component of the velocity of the stone at **A** is 22 m s^{-1}.

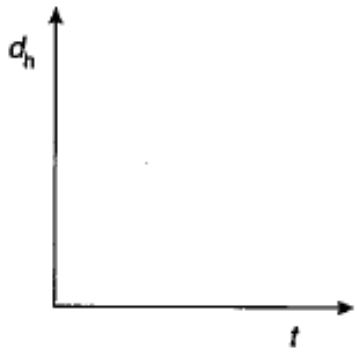
(ii) Calculate the vertical component of the velocity of the stone at **A**.

vertical component..... m s^{-1} [3]

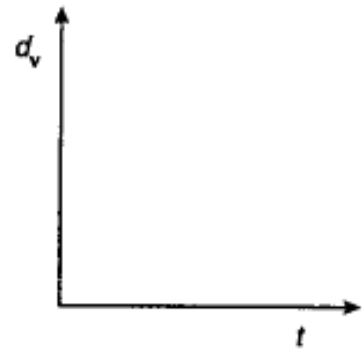
(b) Sketch graphs on the axes to show the horizontal and vertical displacements of the stone from the point of horizontal projection to the point of impact.

Ignore air resistance. Numerical values are not required.

(i) horizontal displacement d_h against time t



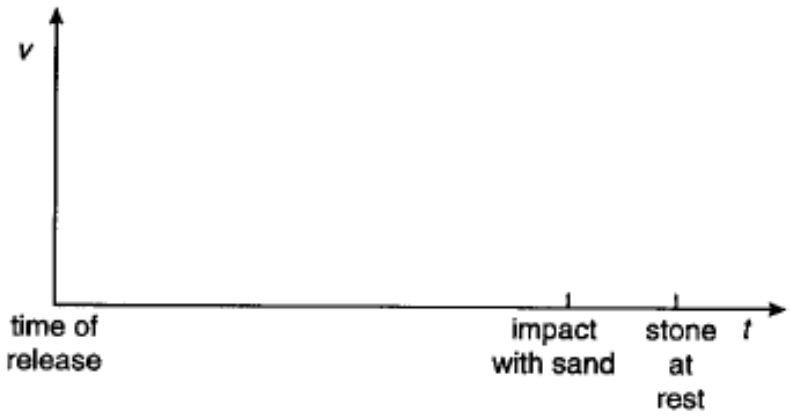
(ii) vertical displacement fallen d_v against time t



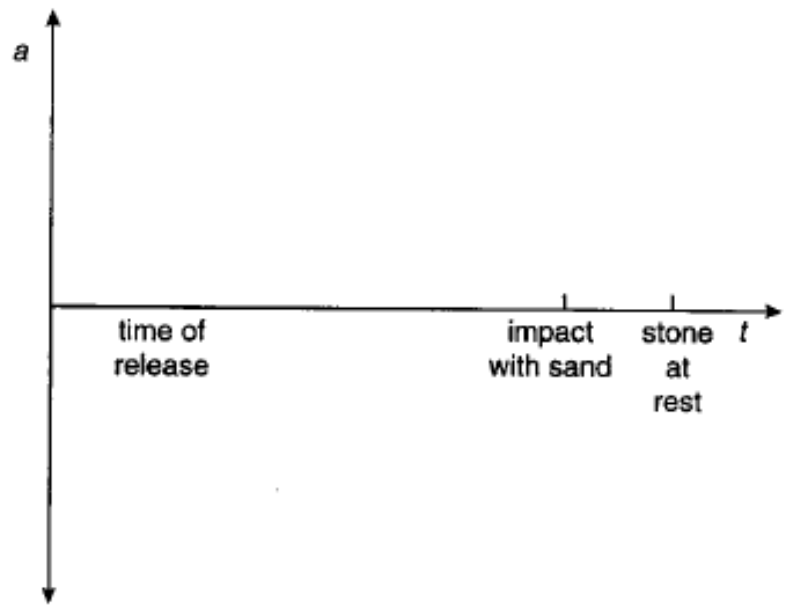
[3]

(c) A second stone is released from rest from the top of the cliff. It falls vertically. Sketch graphs on the axes below to show the velocity v and acceleration a of the stone from the time of release to the time when the stone comes to rest in the sand below. Ignore air resistance. Numerical values are not required.

(i) velocity v against time t



(ii) acceleration a against time t



[5]

(a) In a downhill ski race the total distance between the start and the finish is 1800 m. The total vertical drop is 550 m. The weight of a skier (including his equipment) is 900 N and the time of his descent is 65 s.

(i) Calculate the average speed of the skier for the race.

average speed ..... $m\,s^{-1}$ [2]

(ii) Calculate the loss of gravitational potential energy of the skier.

loss of gravitational potential energy ..... J [2]

(b) The average resistive force acting against the skier is 250 N.

(i) Calculate the work done against this resistive force.

work done ..... J [2]

(ii) If the skier does no work, calculate

1. his kinetic energy as he passes the finish,

kinetic energy ..... J [1]

2. his speed as he passes the finish.

speed ..... $m\,s^{-1}$ [3]

(iii) State whether in practice the final speed of the skier is likely to be greater or less than the value calculated in part **(ii) 2**. Explain your answer.

.....

.....

.....[2]

8

A cable car is used to carry people up a mountain. The mass of the car is 2000 kg and it carries 80 people, of average mass 60 kg. The vertical height travelled is 900 m and the time taken is 5 minutes.

(i) Calculate the gain in gravitational potential energy of the 80 people in the car.

gravitational potential energy gain = ... J [2]

(ii) Calculate the minimum power required by a motor to lift the cable car and its passengers to the top of the mountain.

power = ... unit ..... [3]

(a) A car of total mass 800 kg is travelling along a level road at 25 m s^{-1}. The thinking time of the driver is 0.65 s and the braking distance for the car travelling at this speed is 40 m.

(i) Calculate the overall stopping distance.

stopping distance = ... m

(ii) Calculate the kinetic energy of the car before the brakes are applied.

kinetic energy = ... J

(iii) Calculate the average braking force of the car.

braking force = ... N

[7]

Fig. 6.1 shows a car travelling along a road that has a uniform downhill gradient.

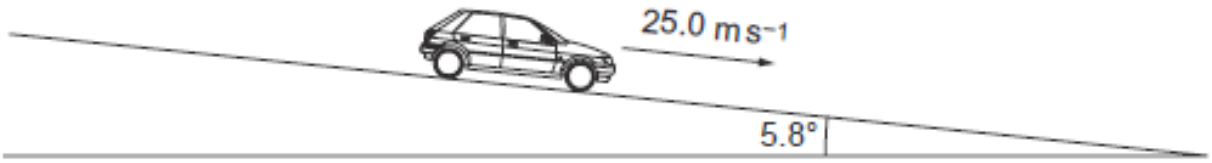


Fig. 6.1

The car has a total mass of 850 kg and is travelling at a constant speed of 25.0 m s^{-1}. The driver sees a hazard ahead and applies the brakes to stop the car. The reaction time of the driver is 0.62 s. When the brakes are applied the braking force is a constant 5520 N.

(i) Calculate the thinking distance.

thinking distance = ... m [2]

(ii) The angle of the road to the horizontal is 5.8°. Show that the deceleration with the brakes applied is 5.50 m s^{-2}.

deceleration = ... m s^{-2} [3]

(iii) Calculate the braking distance of the car.

braking distance = ... m [2]

Fig. 2.1 shows the variation of density of the Earth with **depth** from the surface.

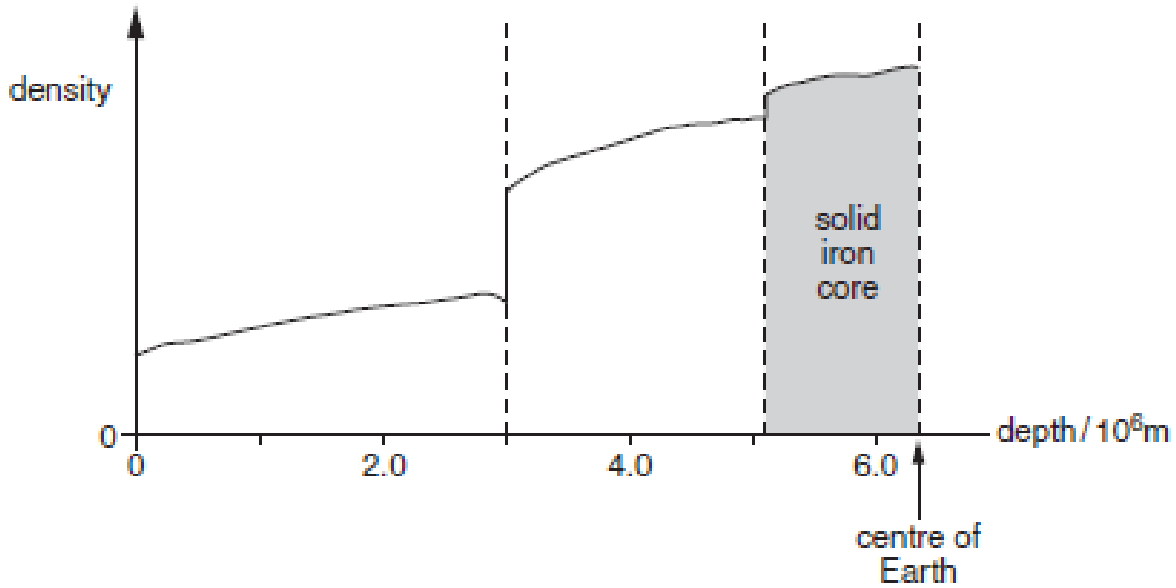


Fig. 2.1

(i) Suggest how Fig. 2.1 shows that the Earth consists of a number of distinct layers.

.....

..... **[1]**

(ii) Geophysicists believe that the central core of the Earth is solid iron. This central core is surrounded by a layer of molten metal. The central core starts at a **depth** of 5.1×10^6 m. The solid iron core accounts for 18% of the mass of the Earth. The mass of the Earth is 6.0×10^{24} kg and its radius is 6.4×10^6 m. Calculate the mean density of the central core of the Earth.

volume of a sphere = $\frac{4}{3}\pi r^3$

density = ... $kg\,m^{-3}$ **[3]**

(a) State two factors that affect the magnitude of the drag force acting on an object falling through air.

1. .....

2. ..... **[2]**

(b) Fig. 4.1 shows a skydiver of total mass 75 kg falling vertically towards the ground.

Fig. 4.1

The air resistance, or drag force, D in newtons (N) acting on the skydiver falling through the air is given by the equation

$$D = 0.3v^2$$

where v is the speed in $m\,s^{-1}$ of the skydiver.

(i) On Fig. 4.1, draw arrows to represent the weight (labelled W) and drag force (labelled D). **[1]**

(ii) Calculate the weight of the skydiver.

weight = ... N **[1]**

(iii) At a particular instant, the speed of the skydiver is $20\,m\,s^{-1}$. Calculate the instantaneous acceleration of the skydiver.

acceleration = ... $m\,s^{-2}$ **[3]**

(iv) State the relationship between the forces *W* and *D* when the skydiver reaches terminal velocity.

.....

..... **[1]**

(v) Determine the terminal velocity of the skydiver.

terminal velocity = ... $m\,s^{-1}$ **[2]**

13

A student conducts an experiment to estimate the acceleration g of free fall by dropping a heavy metal ball from a high wall into a sandpit below. Describe how this student can estimate the acceleration g of free fall. Your description should include:

- the measurements taken
- the instruments used to take the measurements
- how the measurements are used to estimate g
- an explanation of why the value for g is only an estimate

In your answer, you should use appropriate technical terms, spelled correctly.

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